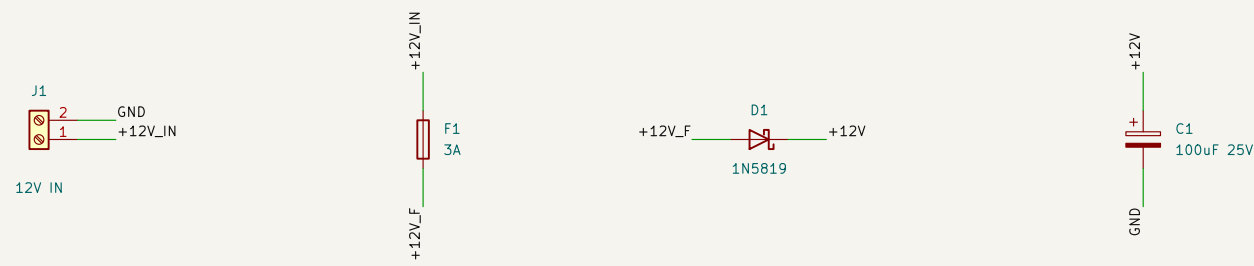
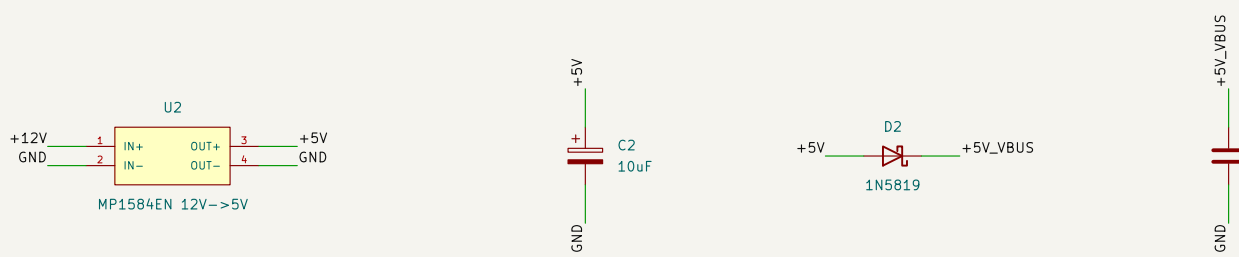


1. 12 V INPUT AND PROTECTION



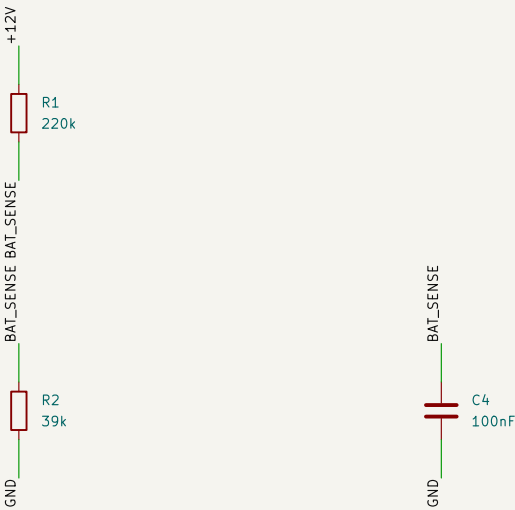
D1 is series reverse-polarity protection, costing ~0.4 V – a 12 V solenoid still operates at 11.6 V. F1 protects the BATTERY, not the board: a shorted valve cable across a lead-acid bank is a fire risk, and nothing else in this design limits that current.

2. 5 V SUPPLY



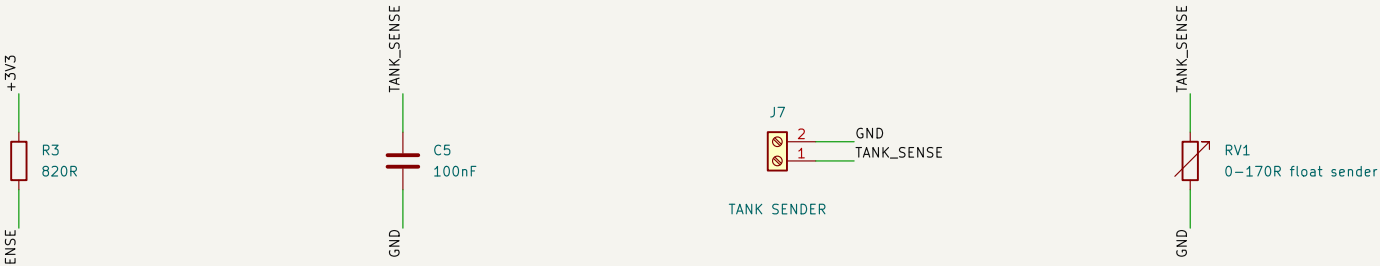
D2 IS NOT OPTIONAL, and the buck setting is part of it. The S2 Mini ties VBUS straight to USB 5 V, so both supplies meet on one node the moment someone plugs in to flash the board. Set the buck to 5.2 V: after D2 the rail is ~4.9 V, just BELOW USB, so when USB is present it wins and nothing is driven back into the host port. D2 also stops USB back-feeding the buck. Do not set the buck above ~5.3 V, or it will push current into the USB host through D2.

3. BATTERY SENSE -> GPIO8



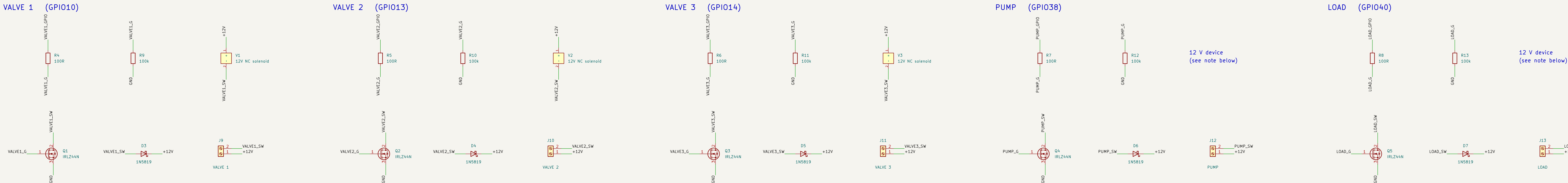
Ratio $(220+39)/39 = 6.641$ – SENSOR_BATTERY_VOLTAGE_DIVIDER must match, and it takes a float. 15 V gives 2.26 V at the pin, inside the S2 ADC's 0..2500 mV linear range; 12.6 V gives 1.90 V. Values are high because a 12 V bank moves slowly, so 49 uA of standing drain is worth more than resolution. C4 is required, not optional: source impedance is $220k || 39k = 33 k$, far above the ADC's 10 k preference; $\tau = 3.3 ms$ against a $10 s$ sample interval. With the lower leg fitted, an absent 12 V rail reads ~0 mV rather than floating – which is why SYSTEM_POWER_BAD_READING_MV=9000 is safe and the old "comment it out on USB" note is gone.

4. TANK LEVEL -> GPIO6



R3 and the sender divide 3V3; OR empty, 170R full -> 0.567 V full. An OPEN circuit pulls the pin to 3V3, which the firmware reports as "no sender fitted" rather than as an empty tank – the distinction OSPIT could not make, and why a missing sender does not block irrigation. The resistances are nominal and were never measured, so calibrate: Tare with the tank empty, then enter 100 with it full. Run the sender on shielded or twisted cable.

8. OUTPUT DRIVERS – five identical low-side channels, all 12 V DC



ACTIVE HIGH, and it is not configurable: Actuator_Digital::act() is a bare digitalWrite(pin, value) with no invert flag anywhere in the class. Most cheap opto-isolated relay boards are ACTIVE LOW and would run every output backwards – do not substitute one without changing the firmware.

The 100k gate pull-downs are a safety part: no decoration. GPIOs float before setup() runs, during a reset, and through a brownout: the pull-down is what holds every valve shut across that window. Valves MUST be NORMALLY CLOSED and must not be latching – loss of 12 V then closes them mechanically, and that, not the firmware, is what guarantees a dead controller cannot leave water running. A latching valve holds its last state and the firmware cannot drive one anyway.

PUMP and LOAD switch the low side of a 12 V device. For a MAINS pump or router, drive an external Sonoff for the frugal-!ot server instead: nothing on this sheet is rated for mains, and mixing mains with low voltage raises the safety bar a long way beyond what this drawing covers.

6. CONTROLLER

